

administration and HR

```
en
conf t
hostname Admin-HR-Switch
```

```
vlan 10 name
Administration
exit vlan
20 name
HR
exit
```

```
interface range fa0/1-10
switchport access vlan 10
exit
```

```
int range fa0/11-20 switchport
access vlan 20
exit
```

```
interface      fa0/24
switchport mode trunk
no shut
exit
```

##Cargo

```
en
conf t
hostname Cargo-Switch
```

```
vlan 50
name Cargo_Operator_Building
exit
```

```
interface range fa0/1-20
switchport access vlan 50
exit
```

```
interface fa0/24
switchport mode trunk
no shut exit
```

##Warehouse

```
en conf
t
hostname Ware-Switch
```

```
vlan 60 name
Warehouse_Building
exit
```

```
interface range fa0/1-20
switchport access vlan 60
exit
```

```
interface fa0/24
switchport mode trunk
no shut exit
```

custom and finance

```
en
conf t hostname Custom-
Finance-Switch
```

```
vlan 30
name Finance
exit
```

```
vlan 40
name Customs exit
```

```
interface range fa0/1-10
switchport access vlan 30 exit
```

```
int range fa0/11-20
switchport access vlan 40
exit
```

```
interface fa0/24
switchport mode trunk
no shut exit
```

security and it

```
en conf t hostname  
Security-IT-Switch
```

```
vlan 70 name  
Security exit
```

```
vlan 80 name  
IT_Department  
exit
```

```
interface range fa0/1-10  
switchport access vlan 70  
exit
```

```
int range fa0/11-20 switchport  
access vlan 80  
exit
```

```
interface fa0/24  
switchport mode trunk  
no shut exit
```

##Data center

```
en  
conf t  
hostname Center-Switch
```

```
vlan 90 name  
Data_Center  
exit
```

```
interface range fa0/1-20  
switchport access vlan 90  
exit
```

```
interface fa0/24  
switchport mode trunk  
no shut exit
```

##CCTV

```
en
conf t hostname CCTV-
Switch
```

```
vlan 120 name
CCTV
exit
```

```
interface range fa0/1-10
switchport access vlan 120
exit
```

```
interface fa0/24
switchport mode trunk
no shut exit
```

##logistic

```
en conf
t
hostname Logistic-Switch
```

```
vlan 130 name
Logistics_Planning exit
```

```
interface range fa0/1-20
switchport access vlan 130
exit
```

```
interface      fa0/24
switchport mode trunk
no shut
exit
```

#guest area

```
en conf
t
```

```
vlan 110 name
Guest_VLAN_110
exit
```

```
int range fa0/2-10 switchport
access vlan 110
exit
```

```
int fa0/24 switchport
mode trunk no shut
exit
```

#R2 router to core

```
en
conf t
hostname R2-admin-hr
```

```
int gig0/0
no shut
exit
```

```
int gig0/1
ip add 10.0.0.6 255.255.255.252
no shut
exit
```

```
int gig0/0.10 encapsulation dot1Q
10 ip add 192.168.10.1
255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
int gig0/0.20 encapsulation dot1Q
20
ip add 192.168.20.1 255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
router ospf 1 network 192.168.10.0
0.0.0.255 area 0 network 192.168.20.0
0.0.0.255 area 0 network 10.0.0.4
0.0.0.3 area 0
exit
```

#R3 router to core

```
en conf t hostname  
R3-cargo
```

```
int gig0/0  
no shut exit
```

```
int gig0/1 ip add 10.0.0.10  
255.255.255.252 no shut exit
```

```
int gig0/0.50 encapsulation dot1Q  
50 ip add 192.168.50.1  
255.255.255.0  
ip helper-address 192.168.90.3  
exit
```

```
router ospf 1 network 192.168.50.0  
0.0.0.255 area 0 network 10.0.0.8  
0.0.0.3 area 0  
exit
```

#R4 router to core

```
en  
conf t  
hostname R4-warehouse
```

```
int gig0/0  
no shut exit
```

```
int gig0/1  
ip add 10.0.0.14 255.255.255.252  
no shut exit
```

```
int gig0/0.60 encapsulation dot1Q  
60 ip add 192.168.60.1  
255.255.255.0  
ip helper-address 192.168.90.3  
exit
```

```
router ospf 1 network 192.168.60.0  
0.0.0.255 area 0 network 10.0.0.12  
0.0.0.3 area 0
```

exit

#R5 router to core

```
en
conf t hostname R5-custom-
finance
```

```
int gig0/0
no shut exit
```

```
int gig0/1 ip add 10.0.0.18
255.255.255.252 no shut
exit
```

```
int gig0/0.30 encapsulation dot1Q
30 ip add 192.168.30.1
255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
int gig0/0.40 encapsulation dot1Q
40 ip add 192.168.40.1
255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
router ospf 1
network 192.168.30.0 0.0.0.255 area 0 network
192.168.40.0 0.0.0.255 area 0 network
10.0.0.16 0.0.0.3 area 0
exit
```

#R6 router to core

```
en conf t hostname R6-
security-it
```

```
int gig0/0
no shut exit
```

```
int gig0/1 ip add 10.0.0.22
255.255.255.252 no shut exit
```

```
int gig0/0.70 encapsulation dot1Q
70 ip add 192.168.70.1
255.255.255.0 ip helper-address
192.168.90.3
exit
```

```
int gig0/0.80 encapsulation dot1Q
80 ip add 192.168.80.1
255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
router ospf 1 network 192.168.70.0
0.0.0.255 area 0 network 192.168.80.0
0.0.0.255 area 0 network 10.0.0.20
0.0.0.3 area 0
exit
```

#R7 router to core

```
en
conf t
hostname R4-datacenter
```

```
int gig0/0 no
shut exit
```

```
int gig0/1 ip add 10.0.0.26
255.255.255.252 no shut exit
```

```
int gig0/0.90 encapsulation dot1Q
90 ip add 192.168.90.1
255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
router ospf 1 network 192.168.90.0
0.0.0.255 area 0 network 10.0.0.24
0.0.0.3 area 0
exit
```

#R8 router to core

```
en conf t hostname
R8-logistics
```

```
int gig0/0
no shut exit
```

```
int gig0/1 ip add 10.0.0.30
255.255.255.252 no shut
exit
```

```
int gig0/0.130 encapsulation dot1Q
130 ip add 192.168.130.1
255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
router ospf 1 network 192.168.130.0
0.0.0.255 area 0 network 10.0.0.28
0.0.0.3 area 0
exit
```

#R9 router to core

```
en
conf t
hostname R9-cctv
```

```
int gig0/0
no shut exit
```

```
int gig0/1 ip add 10.0.0.34
255.255.255.252 no shut exit
```

```
int gig0/0.120 encapsulation dot1Q
120 ip add 192.168.120.1
255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
router ospf 1 network 192.168.120.0
0.0.0.255 area 0 network 10.0.0.32
0.0.0.3 area 0
exit
```

#R10 router to core

```
en conf
t
```

```
int gig0/0 no
shut
exit
```

```
int gig0/1 ip add 10.0.0.70
255.255.255.252 no shut
exit
```

```
int gig0/0.110 encapsulation dot1Q
110 ip add 192.168.110.1
255.255.255.0 ip helper-address
192.168.90.3
exit
```

```
router ospf 1 network 192.168.110.0
0.0.0.255 area 0 network 10.0.0.68
0.0.0.3 area 0 exit
```

##ISP router(R0) -----

```
en
conf t hostname
ISP-RO
```

```
int gig0/0
```

```
ip add 203.0.113.1 255.255.255.252
no shut exit
```

```
ip route 0.0.0.0 0.0.0.0 gig0/0 203.0.113.2 (amr jottorkm ip ache shbgulai ei ip dye replace korbe)
```

##R1 to core -----

```
enable configure
terminal
hostname Edge-R1
```

```
interface GigabitEthernet0/1 ip address
203.0.113.2 255.255.255.252
ip nat outside
no shutdown
exit
```

```
interface GigabitEthernet0/0 ip
address 10.0.0.1 255.255.255.252
ip nat inside no
shutdown exit
```

```
interface Serial0/0/0
ip address 10.0.0.37 255.255.255.252
ip nat inside
no shutdown
exit
```

```
interface Serial0/0/1
ip address 10.0.0.53 255.255.255.252
ip nat inside
no shutdown
exit
```

```
router ospf 1 network 10.0.0.0
0.0.0.3 area 0 network
10.0.0.36 0.0.0.3 area 0 network
10.0.0.52 0.0.0.3 area 0
default-information originate exit
```

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

```
access-list 1 permit 10.0.0.0 0.255.255.255 access-list
1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 int gig0/1 overload
```

##core switch -----

```
en
conf t
hostname Core-Switch
```

```
ip routing
```

```
int fa0/24
no switchport ip add 10.0.0.2
255.255.255.252 no shut
exit
```

```
int fa0/1 no switchport ip add
10.0.0.5 255.255.255.252 no
shut exit
```

```
int fa0/2 no switchport ip add
10.0.0.9 255.255.255.252 no
shut
exit
```

```
int fa0/3 no switchport ip add
10.0.0.13 255.255.255.252 no
shut exit
```

```
int fa0/4 no switchport ip add
10.0.0.17 255.255.255.252 no
shut exit
```

```
int fa0/5 no switchport ip add
10.0.0.21 255.255.255.252 no
shut
exit
```

```
int fa0/6 no switchport ip add
10.0.0.25 255.255.255.252 no
shut
exit
```

```
int fa0/7 no switchport ip add
10.0.0.29 255.255.255.252 no
shut exit
```

```
int fa0/8
no switchport
ip add 10.0.0.33 255.255.255.252
no shut
exit
```

```
int fa0/23 no
switchport
ip add 10.0.0.69 255.255.255.252
no shut
exit
```

```
router ospf 1 network 10.0.0.0
0.255.255.255 area 0 network
192.168.10.0 0.0.0.255 area 0 network
192.168.20.0 0.0.0.255 area 0 network
192.168.50.0 0.0.0.255 area 0 network
192.168.60.0 0.0.0.255 area 0 network
192.168.30.0 0.0.0.255 area 0 network
192.168.40.0 0.0.0.255 area 0 network
192.168.70.0 0.0.0.255 area 0 network
192.168.80.0 0.0.0.255 area 0 network
192.168.90.0 0.0.0.255 area 0 network
192.168.130.0 0.0.0.255 area 0 network
192.168.120.0 0.0.0.255 area 0 network
192.168.110.0 0.0.0.255 area 0 exit
```

EDGE to branch 1:

```
en
conf t int serial0/0/0 ip add
10.0.0.37 255.255.255.252 no
shut exit
```

Main branch 1 router er jonno:

```
en conf t int serial0/0/0 ip add
10.0.0.38 255.255.255.252 no
shut exit
```

```
int gig0/1 ip add 10.0.0.41
255.255.255.252
no shut exit
```

```
int gig0/2 ip add 10.0.0.45
255.255.255.252 no shut exit
```

```
router ospf 1 network 10.0.0.36
0.0.0.3 area 0 network 10.0.0.40
0.0.0.3 area 1 network 10.0.0.44
0.0.0.3 area 1
exit
```

Switch 1:

=====

```
en conf
t vlan
140
name Administration
exit vlan
150
name HR
exit
```

```
int range fa0/1-10
switchport access vlan 140
exit
```

```
int range fa0/11-20
switchport access vlan 150 exit
```

```
int fa0/24 switchport
mode trunk no shut
exit
```

Switch 2:

```
=====
```

```
en conf t
vlan 160
name Finance
exit vlan 170
name Security
exit
```

```
int range fa0/1-10
switchport access vlan 160
exit
```

```
int range fa0/11-20
switchport access vlan 170 exit
```

```
int fa0/24 switchport
mode trunk no shut
exit
```

Router 1:

```
=====
```

```
en conf t
int gig0/0
no shut
```

```
int gig0/1 ip add 10.0.0.42
255.255.255.252
no shut
```

```
int gig0/2 ip add 10.0.0.49
255.255.255.252
no shut exit
```

```
int gig0/0.140 encapsulation dot1Q
140 ip add 192.168.140.1
255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
int gig0/0.150 encapsulation dot1Q
150 ip add 192.168.150.1
255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
router ospf 1 network 10.0.0.40 0.0.0.3
area 1 network 10.0.0.48 0.0.0.3 area 1
network 192.168.140.0 0.0.0.255 area 1
network 192.168.150.0 0.0.0.255 area 1
exit
```

Router 2:

```
en conf t
int gig0/0
no shut
```

```
int gig0/1 ip add 10.0.0.46
255.255.255.252
no shut int gig0/2 ip add
10.0.0.50 255.255.255.252
no shut exit
```

```
int gig0/0.160 encapsulation dot1Q
160 ip add 192.168.160.1
255.255.255.0 ip helper-address
192.168.90.3
exit
```

```
int gig0/0.170 encapsulation dot1Q
170 ip add 192.168.170.1
255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
router ospf 1 network 10.0.0.44 0.0.0.3
area 1 network 10.0.0.48 0.0.0.3 area 1
network 192.168.160.0 0.0.0.255 area 1
network 192.168.170.0 0.0.0.255 area 1
exit
```

EDGE to branch 2:

--

EDGE :

```
en conf t int serial0/0/1 ip add
10.0.0.53 255.255.255.252 no
shut exit
```

Main brunch 2 router r jonno:

```
en conf t int serial0/0/0 ip add
10.0.0.54 255.255.255.252 no
shut
exit
```

```
int gig0/1 ip add 10.0.0.57
255.255.255.252
no shut exit
```

```
int gig0/2 ip add 10.0.0.61
255.255.255.252 no shut
exit
```

```
router ospf 1 network 10.0.0.52
0.0.0.3 area 0 network 10.0.0.56
0.0.0.3 area 2 network 10.0.0.60
0.0.0.3 area 2
exit
```

Switch 1:

=====

```
en conf
t vlan
180
name Administration
exit vlan
190
name HR
exit

int range fa0/1-10
switchport access vlan 180
exit

int range fa0/11-20
switchport access vlan 190 exit

int fa0/24 switchport
mode trunk no shut
exit
```

Switch 2:

=====

```
en conf t
vlan 200
name Finance
exit vlan
210
name
Security
exit

int range fa0/1-10
switchport access vlan 200
exit

int range fa0/11-20
switchport access vlan 210
exit
```

```
int fa0/24 switchport
mode trunk no shut
exit
```

Router 1: =====

```
en conf t
int gig0/0
no shut
```

```
int gig0/1 ip add 10.0.0.58
255.255.255.252 no shut
```

```
int gig0/2 ip add 10.0.0.65
255.255.255.252 no shut exit
```

```
int gig0/0.180 encapsulation dot1Q
180 ip add 192.168.180.1
255.255.255.0
ip helper-address 192.168.90.3
exit
```

```
int gig0/0.190 encapsulation dot1Q
190 ip add 192.168.190.1
255.255.255.0 ip helper-address
192.168.90.3
exit
```

```
router ospf 1 network 10.0.0.56 0.0.0.3
area 2 network 10.0.0.64 0.0.0.3 area 2
network 192.168.180.0 0.0.0.255 area 2
network 192.168.190.0 0.0.0.255 area 2
exit
```

Router 2:

```
en conf t
int gig0/0
no shut
```

```
int gig0/1 ip add 10.0.0.62  
255.255.255.252  
no shut int gig0/2 ip add  
10.0.0.66 255.255.255.252  
no shut exit
```

```
int gig0/0.200 encapsulation dot1Q  
200 ip add 192.168.200.1  
255.255.255.0  
ip helper-address 192.168.90.3  
exit
```

```
int gig0/0.210 encapsulation dot1Q  
210 ip add 192.168.210.1  
255.255.255.0  
ip helper-address 192.168.90.3  
exit
```

```
router ospf 1 network 10.0.0.60 0.0.0.3  
area 2 network 10.0.0.64 0.0.0.3 area 2  
network 192.168.200.0 0.0.0.255 area 2  
network 192.168.210.0 0.0.0.255 area 2  
exit
```